Solution

Equation of ellipse is $x^2+4y^2 = 32$

$$\frac{x^2}{32}+\frac{y^2}{8} = 1$$

$$a^2 = 32,\ b^2 = 8$$

$$a = 4\sqrt{2},\ b = 2\sqrt{2}$$

Equation of tangent at $\theta = \frac{\pi}{4}$ is

$$\frac{x\cos\frac{\pi}{4}}{4\sqrt{2}}+\frac{y\sin\frac{\pi}{4}}{2\sqrt{2}} = 1$$

$$\frac{x}{8}+\frac{y}{4} = 1$$

$$x+2y-8 = 0\,.$$

Equation of normal is $\dfrac{4\sqrt{2}x}{\cos\frac{\pi}{4}}-\dfrac{2\sqrt{2}y}{\sin\frac{\pi}{4}} = 32-8$

That is $8x-4y = 24$

$$2x-y-6 = 0\,.$$

Aliter

$$\text{At},\quad \theta = \frac{\pi}{4}\ ,$$

$$(a\cos\theta,\ b\sin\theta) = \left(4\sqrt{2}\cos\frac{\pi}{4},\ 2\sqrt{2}\sin\frac{\pi}{4}\right)$$

$$= (4,2)$$

$\therefore$ Equation of tangent at $\theta = \frac{\pi}{4}$ is same at $(4,2)$.

Equation of tangent in cartesian form is $\dfrac{xx_1}{a^2}+\dfrac{yy_1}{b^2} = 1$

$$x+2y-8 = 0$$

Slope of tangent is $-\frac{1}{2}$

Slope of normal is 2

Equation of normal is $y-2 = 2(x-4)$

$$y-2x+6 = 0\,.$$

EXERCISE 5.4

1. Find the equations of the two tangents that can be drawn from $(5,2)$ to the ellipse $2x^2+7y^2=14$.
2. Find the equations of tangents to the hyperbola $\frac{x^2}{16}-\frac{y^2}{64}=1$ which are parallel to $10x-3y+9=0$.
3. Show that the line $x-y+4=0$ is a tangent to the ellipse $x^2+3y^2=12$. Also find the coordinates of the point of contact.

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